

Does Codebook Balance Help Generative Multimodal Retrieval? A Controlled Study Across Datasets and Directions

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Abstract. Residual-quantization (RQ) generative retrievers assign every candidate a short discrete semantic ID and decode it directly, replacing nearest-neighbor search; codebook collapse – a few codes absorbing most assignments – is a known failure mode, and balance regularization (an auxiliary loss pushing code usage toward uniform) is the standard remedy. But does more balanced code usage actually improve retrieval? We test this with a controlled ablation of regularization strength on GENIUS, a state-of-the-art RQ-based generative multimodal retriever, across MSCOCO and VisualNews (broad-domain, bidirectional text↔image retrieval) and FashionIQ (fine-grained composed retrieval). We find (1) balance regularization does not consistently improve retrieval: it raises MSCOCO text-to-image Recall@1 significantly (+12.3% relative, $p < 10^{-6}$, verified across 5 seeds and 3 independently retrained tokenizers) but hurts FashionIQ and reverses sign on VisualNews (−53%); (2) its effect differs sharply by direction on identical checkpoints – the same regularization collapses image-to-text Recall from 10.3% to near zero, and a natural per-modality fix rescues neither direction; (3) both effects are already present at the tokenizer level, before decoder training, via a decoder-free probe, and do not transfer consistently across datasets. Absolute Recall in our setup trails published GENIUS numbers, reflecting a matched from-scratch pipeline rather than a state-of-the-art reproduction; our claims concern regularization’s relative effect under this fixed pipeline, not absolute performance.

Keywords: Generative retrieval · Residual quantization · Semantic IDs · Multimodal retrieval · Codebook balance

1 Introduction

Generative retrieval replaces nearest-neighbor search over dense embeddings with autoregressive generation of a discrete identifier that points directly to the target candidate [13]. A growing line of work builds this identifier from a residual quantizer (RQ) [4]: each candidate is assigned a short sequence of discrete codes that a decoder learns to generate directly for a query, retrieving in

time independent of corpus size [12]. Since this code sequence is the only channel through which a candidate’s identity reaches the decoder, codebook quality is a ceiling on the whole system – and RQ codebooks are known to suffer *collapse*, where a few codes absorb most assignments while the rest receive little training signal. The standard remedy is *balance regularization*: an auxiliary loss that pushes code usage toward uniform.

Does more balanced code usage actually lead to better retrieval?

This is not obvious a priori: broader usage should reduce collisions between candidates, but the regularizer pulls the encoder away from the pure-reconstruction optimum, and its effect may depend on task structure – a diverse pool may benefit from spreading code usage, while a pool of near-duplicates may need the encoder to preserve subtle differences any diversity pressure would blur. Despite balance regularization’s routine adoption as a collapse remedy, its causal effect on downstream retrieval Recall is rarely evaluated directly.

We study this question using GENIUS [3], a state-of-the-art RQ-based generative multimodal retriever, as a controlled testbed: we train RQ tokenizers at multiple regularization strengths across three datasets spanning broad-domain bidirectional and fine-grained composed retrieval (Section 3), holding every other pipeline component fixed so regularization strength is the only difference within any comparison. We find:

1. **Balance regularization does not consistently improve retrieval.** It significantly raises MSCOCO text-to-image Recall, but hurts FashionIQ outright and reverses sign on VisualNews.
2. **Its effect differs substantially across retrieval direction.** The identical regularization that helps MSCOCO text-to-image collapses image-to-text Recall on the same checkpoints, and a natural per-modality fix to decouple the two does not rescue either direction.
3. **The degradation is already visible at the tokenizer level and does not consistently transfer across datasets.** A decoder-free probe over the frozen, pre-decoder RQ embeddings shows both effects before any decoder training begins, but neither the direction nor the magnitude of the effect is a stable, portable property.

2 Background and Related Work

Generative retrieval decodes an identifier for the target document instead of searching a dense index [13, 1]. A parallel line builds this identifier from a residual quantizer [4], learning *discrete semantic IDs* instead of hand-built ones – e.g. TIGER [12] for recommendation – though effectiveness degrades sharply with corpus size and task diversity [9]. GENIUS [3] is the multimodal instance: a CLIP-style encoder [10] produces a joint image/text embedding, an RQ quantizes it into a semantic ID, and a T5 [11] decoder generates a target ID via constrained beam search [1] over M-BEIR [15].

RQ codebooks are EMA-fit with no term rewarding even code usage, and are known to suffer *codebook collapse*: a small code subset absorbs most assignments while the rest receive vanishing gradient [8]. The standard remedy adds an entropy-style auxiliary loss alongside reconstruction (*balance regularization*) [17]; an alternative line avoids collapse by construction instead, e.g. FSQ [7]. LETTER [14] adds a diversity loss to RQ-VAE codebooks for generative recommendation – the closest prior use of the intervention we study – as one term in a joint objective for a single-modality identifier, not evaluated for its effect on downstream retrieval or on a codebook serving two target modalities at once, as ours does. None of this prior work evaluates whether balance regularization helps or hurts *retrieval*, or whether its effect is stable across regimes – the gap this paper addresses.

3 Experimental Setup

Testbed. We use GENIUS [3] as testbed throughout: a CLIP-SF (CLIP with score fusion) encoder produces joint image/text embeddings, an 8-level RQ (plus a modality-indicator level, codebook size 4096) quantizes each into a semantic ID, and a T5 decoder generates a query’s target ID via constrained beam search (50 beams) over a trie of valid candidate IDs, so decoding can never emit a nonexistent ID.

Datasets as contrasting regimes. We train and evaluate on MSCOCO [5], VisualNews [6], and FashionIQ [16] independently. MSCOCO and VisualNews are both broad-domain, bidirectional captioning retrieval (text→image, image→text) drawn from different image/text distributions (everyday photography vs. news photography and captions); FashionIQ is composed retrieval (image + modification text → target image) in one narrow domain, its pool dense with visually similar garments. Candidate pools differ substantially in size – MSCOCO 713,679, VisualNews 199,903 (100K image, 99.9K text), FashionIQ 74,380 – so we never compare absolute Recall across datasets, only relative effects within each. Pool size is also not what drives the dataset-dependence we report: re-running MSCOCO’s ID-structure diagnostics on a subsample size-matched to FashionIQ’s pool (74,380 candidates) still yields 2.1–4.0× FashionIQ’s codebook utilization at every λ , with far steeper collision growth. For VisualNews, no official M-BEIR held-out test partition exists in our downloaded data, so we repurpose the union pool’s validation split (never seen during training) as the test set – a disclosed deviation from the official-test-split convention used for MSCOCO and FashionIQ.

Pipeline and controls. For each dataset we fix the CLIP-SF encoder, extract candidate embeddings once, then train RQ tokenizers differing only in balance-regularization strength λ – four points $\{0, 0.3, 1.0, 3.0\}$ (vanilla/weak/medium/strong) for MSCOCO and FashionIQ, two points $\{0, 3.0\}$ for VisualNews – holding architecture, level count, codebook size, decoder, and decoding settings fixed. Each tokenizer’s IDs retrain the same decoder from scratch. Headline Recall numbers

Table 1. Recall@1 by dataset and balance strength λ , mean $\pm$ std across seeds. Dir. A = text $\rightarrow$ image (MSCOCO, VisualNews) / (image,text) $\rightarrow$ image (FashionIQ); Dir. B = image $\rightarrow$ text (MSCOCO, VisualNews only). MSCOCO vanilla/strong: $n=5$ seeds; all other rows: $n=3$ seeds. Both MSCOCO’s and VisualNews’s vanilla-vs-strong Dir. A differences are significant under Welch’s t -test (Section 4).

Dataset	λ	Dir. A R@1	Dir. B R@1
MSCOCO	vanilla (0.0)	20.78 ± 0.06	10.30 ± 0.67
MSCOCO	weak (0.3)	14.13 ± 0.53	0.05 ± 0.02
MSCOCO	medium (1.0)	10.36 ± 0.69	0.03 ± 0.01
MSCOCO	strong (3.0)	23.33 ± 0.11	0.03 ± 0.02
VisualNews	vanilla (0.0)	20.07 ± 0.11	2.88 ± 0.05
VisualNews	strong (3.0)	9.43 ± 0.08	0.01 ± 0.00
FashionIQ	vanilla (0.0)	0.61 ± 0.10	–
FashionIQ	weak (0.3)	0.27 ± 0.06	–
FashionIQ	medium (1.0)	0.21 ± 0.09	–
FashionIQ	strong (3.0)	0.25 ± 0.08	–

are means over multiple decoder-training seeds per variant (5 for MSCOCO vanilla/strong, 3 elsewhere); MSCOCO’s headline gain is further verified across 3 independently retrained RQ tokenizers (Section 4), not just decoder seeds.

Scope: relative effects under a fixed, non-SOTA-reproducing pipeline. Because the officially released GENIUS decoder is not public, we pair the officially released Stage-1 RQ checkpoint with our own decoder trained on the full ~ 1.3 M-query, 16-task M-BEIR union, and compare against the published paper’s numbers: FashionIQ lands above the published figure (18.3–18.6% R@10 vs. 13.1%, no reranking) while MSCOCO T $\rightarrow$ I R@1 lands far below it (8.1% vs. 40.1%). We do not read the FashionIQ result as proof the pipeline is faithful – overshooting on one dataset and undershooting on another is itself a reason for caution – only as evidence that the gap is not a uniform implementation failure. The MSCOCO shortfall is consistent with capacity dilution across 16 joint tasks [9], plus our lack of validation-based checkpoint selection and hard negatives – training-recipe gaps we disclose rather than close further. This union-trained reproduction is a separate regime from every λ -sweep comparison in this paper: each sweep variant instead trains a *fresh* RQ tokenizer and decoder from scratch, per dataset, with λ as the only difference, and is never exposed to the union-training gap above. We therefore report the effect of regularization *relative to a matched vanilla baseline under an identical from-scratch pipeline*, not GENIUS’s absolute state-of-the-art performance, and we do not pursue closing the reproduction gap further within this study.

4 Finding 1: Balance Regularization Does Not Consistently Improve Retrieval

We report each dataset’s primary retrieval direction as Dir. A – text→image (T→I) for MSCOCO and VisualNews, composed (image,text)→image for FashionIQ – and, where the dataset is bidirectional, image→text (I→T) as Dir. B. Table 1 shows no consistent effect of balance regularization on Dir. A Recall across datasets. On MSCOCO, strong regularization is significantly better than vanilla: $23.33 \pm 0.11\%$ vs. $20.78 \pm 0.06\%$ ($n=5$ seeds each), non-overlapping ± 1 -std bands, $p < 10^{-6}$ (Welch’s t -test), a +12.3% relative gain – and the gain holds across 3 independently retrained RQ tokenizers, not just decoder seeds: vanilla mean $21.16 \pm 0.47\%$ vs. strong mean $24.32 \pm 0.86\%$, every strong tokenizer beating every vanilla tokenizer. (Absolute values here are well below published GENIUS numbers – see Section 3 – reflecting this study’s matched, from-scratch pipeline rather than a state-of-the-art reproduction.) On VisualNews, the same strong regularization instead *halves* Recall ($20.07\% \rightarrow 9.43\%$, -53% relative; $n=3$ seeds each, Welch’s t -test, $p < 10^{-7}$) – a reversal of sign on a second broad-domain, bidirectional corpus. On FashionIQ, vanilla wins outright at every regularized λ ; the ranking *among* regularized strengths is not seed-stable (all values sub-1% R@1), so we report only that any regularization hurts, not a specific optimal λ .

This gain is also direction-specific – the same MSCOCO checkpoints that improve on Dir. A collapse on Dir. B (Table 1, Section 5) – and its sign even reverses on a second broad-domain, bidirectional dataset (Section 6); we examine both next.

5 Finding 2: The Effect Differs Substantially Across Retrieval Direction

On MSCOCO, the identical checkpoints tell opposite stories depending on direction (Table 1). Image-to-text (I→T, Dir. B) is unambiguous: vanilla is the *only* variant with working I→T retrieval (10.30% mean, $n=5$ seeds, std 0.67); every regularized variant collapses to $\leq 0.05\%$ mean, seed-independently, with std an order of magnitude smaller than vanilla’s own. This holds even at weak regularization, where text-to-image (T→I) is far less affected – the two directions do not degrade together, or at comparable rates.

A natural fix – decoupling regularization per target modality – fails architecturally. One might hypothesize a per-direction loss weight ($\lambda_{\text{img}}/\lambda_{\text{txt}}$ in place of one shared λ) could rescue both directions at once. We tested three settings on MSCOCO (3-seed means $\pm$ std): image-only ($\lambda_{\text{img}}=3, \lambda_{\text{txt}}=0$), image-dominant (3/0.3), and text-only (0/3). It does not work. I→T stays collapsed in all three ($0.12 \pm 0.04\%$, $0.13 \pm 0.05\%$, $0.03 \pm 0.01\%$), including the exact-zero-text-weight setting; the text-only setting collapses I→T as completely as regularizing both sides does. T→I is *worse than vanilla* in all three ($7.87 \pm 0.12\%$, $5.56 \pm 0.13\%$, $6.53 \pm 0.08\%$ vs. vanilla’s 20.78%) – worse than either uniform baseline, not

Table 2. Localizing both effects to the tokenizer (MSCOCO). The dense probe is cosine similarity over the frozen RQ’s quantized embedding – no decoder, no beam search, no trie – run on the single Stage-1 checkpoint per variant shared by all decoder seeds; “real” columns repeat Table 1’s multi-seed pipeline means for comparison.

Variant	Dense T→I	Dense I→T	Real T→I	Real I→T
vanilla	26.65%	7.52%	20.78%	10.30%
weak	17.32%	0.04%	14.13%	0.05%
medium	12.87%	0.00%	10.36%	0.03%
strong	24.23%	0.02%	23.33%	0.03%

merely worse than strong. This is not a tuning failure: image and text candidates share one encoder forward pass and one EMA-updated codebook at every RQ level, so a per-row loss weight has no mechanism to produce a per-modality effect. This is consistent with a direct measurement of the shared structure: under image-only regularization, text-side codes are visibly restructured despite exactly zero text-side loss weight (566 vs. 946 unique level-1 code prefixes relative to vanilla), i.e. image-side regularization reaches text-side code structure through the shared codebook.

Section 6 shows both this collapse and its resistance to per-modality decoupling already exist before decoder training begins, at the RQ tokenizer level.

6 Finding 3: The Effect Originates at the Tokenizer Level and Does Not Transfer Consistently

Localizing both effects to the tokenizer, before decoder training. Table 2 probes the frozen RQ embeddings directly. On MSCOCO, I→T *already* collapses in this dense probe for weak/medium/strong ($\leq 0.04\%$, matching the real pipeline’s band), while vanilla’s dense I→T (7.52%) stays close to its real-pipeline number (10.30% mean); T→I retains substantial, rank-correlated dense recall across all four variants. Regularization degrades the RQ embedding space’s ability to discriminate text candidates for image queries *before decoder training begins* – which also explains why per-modality decoupling (Section 5) cannot fix it: the damage is already encoded in the tokenizer, a place a decoder-side loss weight cannot reach.

VisualNews: the collapse generalizes, the gain reverses – both localize the same way. The same probe on VisualNews’s vanilla/strong checkpoints shows both effects are tokenizer-level there too: dense I→T falls 2.67% → 0.00% (matching the real pipeline’s 2.88% → 0.01%), and dense T→I falls 20.22% → 10.59% ($\sim 48\%$ relative, tracking the real pipeline’s $\sim 53\%$ drop). Neither direction is decoder-introduced on either dataset. One caveat on rigor: as on MSCOCO, the dense probe uses a single frozen Stage-1 checkpoint per variant, so it carries no seed variance of its own; the real-pipeline Recall it corroborates (Table 1) is a 3-seed mean, and the T→I reversal it tracks is itself significant (Section 4).

We do not report a significance test for the I→T collapse on either dataset: the regularized arm sits at the Recall floor (identical across all three seeds on VisualNews, and within 0.00–0.04% on MSCOCO), where a parametric test on a bounded metric compressed against zero would manufacture an impressive p -value rather than earn one. The effect size speaks for itself.

No structural metric predicts Recall consistently across datasets. Correlating codebook utilization, collision rate, and reconstruction quality against Recall across MSCOCO’s and FashionIQ’s four λ points ($n=4$, Pearson r , descriptive) sharpens this: on FashionIQ, structural metrics track Recall almost perfectly and in the conventional direction (collision $r=-0.99$, utilization $r=+0.97$, reconstruction MSE $r=-0.94$), so the familiar “better reconstruction → better retrieval” story holds there. On MSCOCO the same metrics are far weaker and not even monotonic: collision rate carries a positive Pearson coefficient ($r=+0.47$) only because the best-performing configuration also has the highest collision rate, while the other three variants trend the opposite way – its rank correlation is near-null (Spearman $\rho=+0.20$). We therefore do not claim a positive collision–Recall relationship on MSCOCO. We claim the weaker but more damaging thing: no structural metric we measured predicts Recall consistently in sign, strength, or monotonicity across the two regimes, so structural ID metrics are, at best, incomplete and dataset-dependent proxies for retrieval quality.

Together, Findings 1–3 show balance regularization’s effect is neither absent nor uniform: it is real, direction-specific, tokenizer-level, and dataset-dependent in sign – properties a practitioner cannot infer from a single dataset or direction.

7 Limitations and Conclusion

Limitations. As noted in Section 3, absolute Recall in our from-scratch pipeline trails published GENIUS numbers; this sits on exactly the dataset and direction carrying our one clear positive result (MSCOCO Dir. A), so we cannot rule out a different effect size in a fully converged, published-parity regime – a limitation we regard as consistent with, rather than contradicting, this paper’s central claim that regularization’s effect is not a stable, portable property of the method. Our significance tests certify robustness to decoder-training-seed noise, not to every source of variance: MSCOCO’s gain is additionally verified across 3 independently retrained Stage-1 tokenizers, but VisualNews’s reversal was checked only across decoder seeds on a single tokenizer per variant, so tokenizer-initialization variance is unmeasured there. Section 6’s correlations use four λ points per dataset; the divergence in their behaviour between datasets is the finding, not $n=4$ significance. We localize the image→text collapse to the tokenizer but do not fully explain it: text candidates start more anisotropic than image candidates in the fused embedding space (mean pairwise cosine similarity 0.842 vs. 0.675) [2], which is consistent with per-sample balance pressure shattering an already-narrow text cluster first, but does not explain why the collapse persists at strong λ once code entropy partially recovers – a full mechanistic account is left to future work. Retrieval-tie-breaking within collision-heavy codes

does not inflate MSCOCO’s headline gain (verified against a random and an adversarial tie-break baseline, both within 0.05 points of the reported value). Our three datasets are contrasting exemplars, not a sweep – the optimal λ for one dataset need not transfer to a further regime, which is itself part of what we report.

Conclusion. Balance regularization is neither useless nor universally beneficial for RQ-based generative multimodal retrieval: it significantly helps MSCOCO text-to-image Recall while destroying image-to-text Recall on the same checkpoints, substantially hurts VisualNews’s text-to-image Recall, and hurts FashionIQ outright. Decoupling regularization per target modality does not rescue either direction, because image and text candidates share one encoder and codebook. Both the direction-asymmetry and the dataset-dependence are already present at the RQ tokenizer level, before any decoder is trained. Practitioners should not adopt balance regularization by default without validating its effect per retrieval direction and per dataset.

Code and data. Code, configurations, and per-seed results for every variant reported here will be released as a public repository at acceptance.

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